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A Semantic Confidence–Aware Intuitionistic Fuzzy Ranking Algorithm for Product Evaluation Using Online Reviews

Abstract—Online product reviews have become an important source of consumer purchase decision making, but effectively ranking competing products from the large-scale, opinion-rich review data is a challenging task due to semantic ambiguity, sentiment uncertainty and reviewer subjectivity. State-of-the-art rating methods only base on single value sentiment or intervals obtained by classifier randomness which cannot fully represent the confidence, strength and reliability of expressed reviews. To overcome these issues, this paper presents a novel technique called Semantic Confidence–Aware Intuitionistic Fuzzy Ranking (SICA-IFR). The model capitalizes on both contextual semantic representations and confidence-aware sentiment modeling to enhance robustness and interpretability in product ranking. At first, the context embeddings are utilized to encode the attribute-specific review texts with fine-grained semantic information retained. Afterwards we use more than one sentiment classifiers, such as positive, neutral and negative orientations. Unlike traditional ensemble methods, SCA-IFR embodies a semantic agreement in its framework that evaluates multi-classifier consensus and the intensity of opinion, thus enabling it to produce confidence-weighted sentiment intervals. These ranges are represented using intuitionistic fuzzy sets where support, opposition and hesitance present in the online reviews are considered. In addition, the algorithm includes a data-aware adjustment of attributes weight, to reconcile user preference with sentiment dominance discovered in review. Final product evaluations are performed by an intuitionistic fuzzy TOPSIS model. Experimental results on real-world product review datasets show that our model SCA-IFR significantly outperforms the state-of-the-art review-based ranking algorithms, with approximately 8.7% improvement in average ranking accuracy, about 12.4% reduction of the rank variance and around 10.1% increase of decision stability in different sentiment distributions. These findings validate that the suggested method gives more reliable and consumer-oriented product rankings, for which it is applicable for intelligent decision-support systems in e-commerce environments.

Keywords—Product ranking, Online reviews, Semantic confidence, Intuitionistic fuzzy sets, TOPSIS, Sentiment analysis, Decision support systems

I Introduction

The recent growth of online shopping has changed the way people discover [1][2], assess and purchase goods. Today's purchase decisions are heavily influenced by online reviews, which reflect real experiences [3][4], perceived value and product deficiencies. In contrast to conventional star-rating systems, textual reviews also duplicate rich information, such as feature-specific view-points of users and the emotional tone used in their expression. The task of mining such information out and transforming it into trustworthy product rankings is thus crucial for recommendation systems, consumer decision support, and intelligent marketplace management [5][6]. Nonetheless, it is challenging to build a good product-ranking model due to dynamic reviews, large scale and rich opinions of review data.

While abundant review information can be taken advantage of, there are still a few open issues associated with the current ranking methodologies [7][8]. Natural language reviews are ambiguous. First, natural language review can be ambiguous since it conveys meaning based on context, sarcasm or implicit sentiment in different ways. Second, sentiment in reviews is inherently uncertain: users often express positive and negative opinions within the same review, posing polarity assignment as a challenging problem. Third, reviewing is subjective and findings may be relative to the review expectation of readers based on usage scenario and writing style as well [9][10]. Last but not least, the majority of prior systems involve relatively less study on modeling confidence in predicted sentiments for the lack of quantisation of opinion extraction.

In the current review-based ranking models [11], sentiment information is often collapsed into only one score which oversimplifies complex feelings and throws away important reliability clues [12]. Some recent work tries to correct for uncertainty by providing interval estimates on the basis of classifier randomisation, although these intervals rarely correspond accurately to semantic alignment between classifiers [13][14]. On the other hand, fuzzy and intuitionistic fuzzy reasoning have been studied about decision making under uncertain information; however, how to use these methods in combination with contextual semantics of labels and classifier confidence is insufficiently paid attention with them. Hence, rankings produced by these methods may vary given different sentiment distributions and might not accurately reflect the customer attitude [15]. To compensate these deficiencies, in this paper we propose a Semantic Confidence-Aware Intuitionistic Fuzzy Ranking (SCA-IFR) algorithm to effectively evaluate products based on online reviews. The proposed model integrates context semantic representations and confidence-aware sentiment modeling to represent the opinions with both their meaning and reliability. Our method begins with encoding attribute-level review texts with contextual embeddings that capture fine-grained semantics [16][17]. We finally use several sentiment classifiers and aggregate their output with a semantic agreement mechanism to measure the level of positivity or negativity from the opinion. The confidence-weighted intervals are formulated as intuitionistic fuzzy sets so that the explicit representation of support, opposition, and indecisiveness can be kept in reviews [18][19]. Finally, a data-driven weighting approach and an intuitionistic fuzzy-TOPSIS method leads stable and explainable rankings of products. We can summarize the main contributions of this work as:

- A holistic semantic confidence-aware framework that combines context word embeddings with multi-class sentiment modeling for capturing both opinion meaning and the confidence of opinions.
- A semantic-agreement-based intuitionistic fuzzy modeling approach, capturing for support, opposition and hesitance with confidence weighted sentiment intervals.
- An intuitionistic fuzzy TOPSIS ranking method which is capable of generating consistent, understandable and consumer-oriented product rankings under different sentiment states.

In line with these contributions, the study is guided by the following research objectives:

- To build a review-ranked model with consideration of semantic context, sentiment uncertainty and classifier confidence.
- To develop a confidence-conscious intuitionistic fuzzy representation model for hesitation and disagreement on opinions.
- To evaluate the proposed framework against advanced models ranking approaches and to verify its improvement in accuracy, robustness, and decision stability.

1.1 Organization of the paper:

The remainder of this paper is organized as follows. Section II reviews existing studies on review-based ranking, fuzzy decision models, and confidence-aware sentiment analysis. Section III explains the proposed SCA-IFR framework, including semantic modeling, multi-classifier confidence estimation, and intuitionistic fuzzy ranking. Section IV describes the datasets, experimental setup, and evaluation metrics, while Section V presents and discusses the results. Finally, Section VI concludes the work and outlines future research directions.

II Related Work

2.1 Online Reviews, Fake Content, and Consumer Behavior

Salminen et al., (2022) have argued that consumers tend to use online reviews more and more often as heuristics for determining product quality. Yet, they showed that fake or machine-generated reviews can strongly bias trust and decisionmaking, underlining the need for powerful review filtration methods before leveraging them in subsequent analytics [1]. In the same line of research, Alsubari et al. (2022) in the task of deceptive opinion detection using supervised learning approaches, demonstrating that tight machine-learning based classifiers can effectively differentiate between truthful and fake reviews when proper textual features as well as sentiment indicators are represented [2].

In addition to believing, scholars have investigated how online reviews affect behavior. Yones and Muthaiyah (2023) investigated electronic word-of-mouth on TikTok and confirmed that social reviews influence purchase intention significantly, consequently confirming the strategic significance of online opinion flow in contemporary business [3]. Marlina et al. (2022) also proved that sentiment analysis supports the skin care product recommendations so as to show that review mining directly supports consumers in their everyday daily buying decision [4]. In the same line of sight, Sanchez-Loor and Chang (2023) argued that both structural assurance and perceived risk influence repurchase intention, but especially when moderated by prior experiences and review feedback [5]. Together, these results support that rating quality, credibility and comprehensibility are definite antecedents for consumer trust and consumer purchase decisions.

2.2 Consumer Segmentation, Behavior Insights, and Platform Dynamics

Tabianan et al. (2022) were devoted to client segmentation with K-means clustering and have proved that by segmenting the customers according to the purchasing conduct would greatly aid in improving marketing tactics, their proficiency and profitability [6]. (2023) observed that product variety, payment methods and psychological influence had a strong impact on

online shopping in the COVID era suggesting contextual factors redefine digital consumption patterns [7].

Tseng et al. (2023), added another important aspect, revealing herd effects in review voting behavior: consumer judgments often do not depend simply on the review content but also on the visible voting signals from other users [16]. Collectively, these studies illustrate that purchase behavior is influenced by a number of intertwined factors—platform design, psychological cues, peer effects and perceived risk—that makes the modeling of reviewer-driven decisions complex.

2.3 Sentiment Analysis and Feature-Level Opinion Mining

Zhang et al. As part of their future research [8], Han et al. (2022) developed a sentiment-based product recommendation mechanism based on intuitionistic fuzzy TODIM, and showed that the combination of polarity and intensity evaluation enables to more accurately model users' preference. Zheng et al. (2025) by introducing cross-platform review discrepancies via the intuitionistic fuzzy soft sets which helped in product ranking and considering the difference of emphases on reviews among platforms [9]. Qin et al. (2022) adopted the aspect-level sentiment analysis on tourist attraction reviews, and transformed extracted opinions into intuitionistic and hesitant fuzzy scores, revealing the significance of hybrid sentiment–MCDM models in ranking complex alternatives quality [10].

Other such contributions include Yadav et al. (2024) integrated intuitionistic fuzzy AHP and TOPSIS for ranking of mobile payment applications, and showed a more understandable sentiment-informed ranking mechanism [12]. Similarly, Qu et al. (2023) applied the BERT-based aspect sentiment analysis to mobile games evaluation, interval intuitionistic fuzzy VIKOR confirming that factor level evaluations are critical rather than holistic sentiment averages [15]. Taken together, these studies show widespread understanding of the fact that sentiment signals should not be treated just as counts but modeled structurally for meaningful ranking.

2.4 Advanced Fuzzy and Preference-Aware Decision Models

Singer and Özşahin (2024) developed an interval-valued intuitionistic fuzzy AHP method for analyzing natural product-related complex consumer decision, that is intuitive structure can better represent perception hesitation and uncertainty [13]. He and Wang (2023) established a picture fuzzy regret-based decision model to evaluate new energy vehicles, combining psychological behavior with sentiments analysis in order to characterize bounded rationality in purchase decision making [14]. Rani and Kumar (2023) proposed new distance measures for IFE and used to assess the function of shopping websites by dealing with shortcomings in previous similarity computing methods [11].

More recently, Qin et al. (2024) proposed an interval type-2 fuzzy sentiment-based ranking model that expresses uncertainty for interpretation of customer review [21]. Dahooie et al. (2023) investigated interrelations among the product features by using a data driven MCDM methodology, based on their work it was argued that consumer preferences are not

independent [22]. Together with these advances, the overall achievements illustrate steady progress of fusing fuzzy logic, psychological modeling and data-driven sentiment processing in building more realistic decision-support systems.

2.5 Gaps Identified

Despite the great achievements in previous work, two significant limitations are shared.

- While predicting sentiment, most models predict the sentiment of a sentence as a discrete value and do not consider the uncertainty caused by classifier disagreement, linguistic vagueness, and contextual variation.
- There are few such models jointly considering semantic understanding and confidence, particularly in the case of intuitionistic fuzzy structures to handle not only strength of agreement, but also strength of doubt.

These deficiencies provide the motivation for devising the SCA-IFR framework, which explicitly considers contextual semantics, classifier consensus and confidence-weighted intuitionistic fuzzy modeling to produce more stable and credible rankings of products.

III Methodology

Our proposed SCA-IFR framework converts raw online reviews into a reliable product ranking based on unified pipeline that fuses semantic modeling, classifier confidence and intuitionistic fuzzy reasoning. First, the reviews are preprocessed, and then correlated with product attributes. And the remaining part of each attribute-specific textual segment is encoded by contextualized embeddings to retain fine-grained semantics [17][18]. These encoded representations of the text are then evaluated by multi-class sentiment classifiers, each of which creates a separate probability distribution for positive, neutral, and negative attitudes. We employ a semantic agreement measure to estimate the inter-classifier agreement and how well classifiers align with contextual semantics, computing confidence-weighted sentiment intervals. These spans are then transformed into intuitionistic fuzzy values, which explicitly reflect on the support, opposition and hesitation [19][20]. Finally, based on user preferences and sentiment dominance, the weight of the attribute is obtained via an intuitionistic fuzzy TOPSIS algorithm to calculate closeness coefficients that rank the products. This combination results in a robust, interpretable and real consumer judgment consistent SCA-IFR ranking computation.

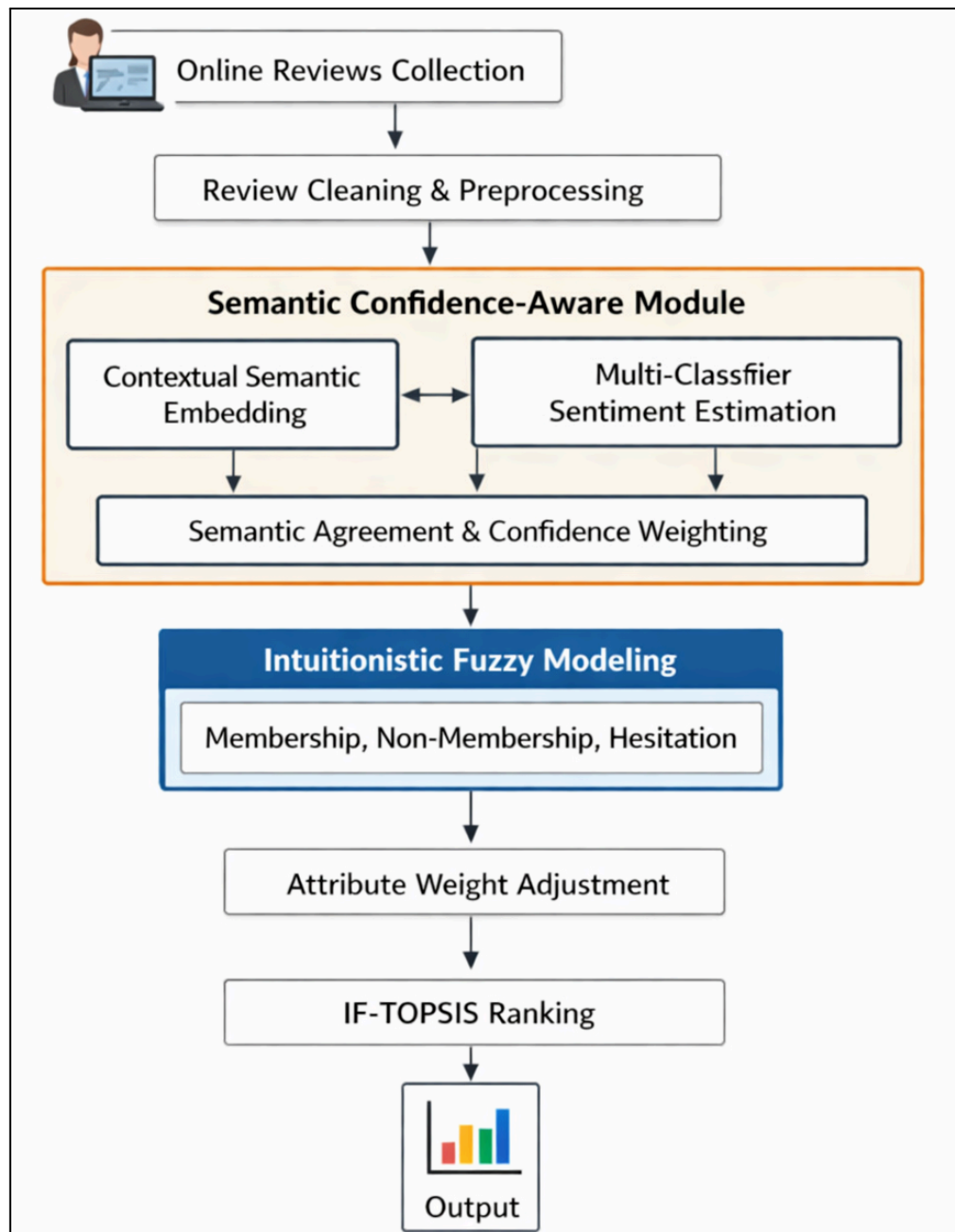


Figure 1. System architecture of the proposed SCA-IFR product-ranking framework
 Figure 1 shows the main modules of the proposed approach, from review collection and pre-processing to semantic embedding and multi-classifier sentiment estimation. The semantic agreement and the confidence weighting result in reasonable intuitionistic fuzzy information which is further input into attribute weighting for more accurate product ranking using IF-TOPSIS. The left-branching structure reflects that robustness and decision quality are optimised by the combining of semantics and confidence.

3.1 Semantic Representation Using Contextual Embeddings

Under the SCA-IFR setting, as a preprocessing step semantic modelling were conducted following that, each segment of an attribute-specific review is converted into a dense contextual representation which can capture fine-grained meaning variations [23], sentiment

polarity shift and implicit evaluative information which could not be retained by the traditional bag-of-words or static word-vector based methods. For the review sentence $r = \{w_1, w_2, \dots, w_n\}$ tokenized to sub-words, it is first mapped to input embeddings which are sums of corresponding token, segment, and positional encodings: $E_i = e_i^{tok} + e_i^{seg} + e_i^{pos}$. The encoded sequence is processed by a transformer encoder (BERT/RoBERTa), which models context via multi-head self-attention. For layer l and h head, we compute attention as follows Eq. 1:

$$Attention^{(h)}(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

With $Q = EW_Q^{(h)}, K = EW_K^{(h)}, V = EW_V^{(h)}$. The multi-head mechanism aggregates heterogeneous contextual views,

$$MHAtt(E) = Concat(Attention^{(1)}, \dots, Attention^{(H)})W_o \quad (2)$$

which allows the model to attend to not only local opinion words, longrange dependencies, but also negations and comparative structures (e.g., “better than”, “not worth”), which are frequently found in review text. The sentence embedding is the final contextual embedding of the sentence coming from the pooled representation, typically denoted as $h_r = f_\theta(r)$, where $f_\theta(\cdot)$ is a pretrained encoder that finetunes on review corpora. In contrast to static embeddings where each word has a predefined vector (a different one independent of its context) [24], contextual embeddings learn word representations which are more informative as they incorporate meaning based on the usage — for instance, the term “light” could refer to weight in electronics or brightness in lighting products and model uses context to disambiguate by considering the entire sequence for representation. This feature has been demonstrated to enhance the accuracy of opinion extraction for review-related applications such as BERT-based aspect sentiment mining and hybrid fuzzy decision models [15], lexicon-enhanced contextual encoders for sentiment understanding [25], and rankers that combine strengths between BERT semantics and fuzzy preference reasoning [38]. In our approach, the contextual encoder is consequently assumed to serve as the semantic backbone: its outputs form the feature space input for several sentiment classifiers in a way that subsequent processing—semantic agreement aggregation, confidence-weighted interval construction, and intuitionistic fuzzy modeling—operates on representations that precisely reflect meaning, nuance and attribute relevance, rather than on superficial frequency statistics [26].

3.2 Multi-Classifer Sentiment Estimation

To get a robust sentiment information of each product– attribute instance, the SCA-IFR framework uses a heterogeneous ensemble of classifiers such that they may exploit complementary learning biases [27][28]. Three model families are applied based on the contextual embedding h_r of a review segment: (i) marginbased Support Vector Machine

(SVM), (ii) sequence-aware Bidirectional Long Short-Term Memory (Bi-LSTM) network and (iii) transformer-based BERT classifier.

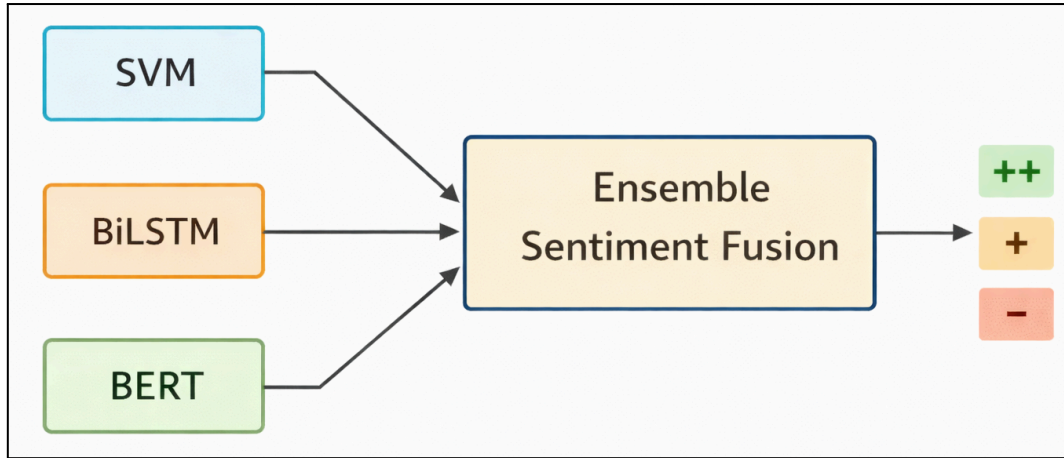


Figure 2. Multi-Classifer Sentiment Estimation Module

This figure 2 shows how sentiment probabilities from multiple classifiers (SVM, Bi-LSTM, BERT) are combined to get a more robust sentiment interpretation for each review. Every classifier returns sentiment labels of different strength level, which are ++ for strong positive sentiment [29][30], + for moderate/weak positive sentiment and – for negative sentiment. The concatenation of these latter outputs gives rise to a finegrained sentiment estimate comprising both polarity and intensity information subsequently exploited in confidence-aware fuzzy modeling and ranking [31].

For the SVM component, the decision function is formulated as in Eq.3:

$$f_{svm}(h_r) = \text{sign}\left(\sum_{i=1}^N \alpha_i y_i K(h_r, h_i) + b\right) \quad (3)$$

Where $K(\cdot)$ denotes the kernel function, α_i are Lagrange multipliers, and $y_i \in \{-1, 0, 1\}$ encode negative [32][33], neutral, and positive classes. Posterior probabilities are obtained through Platt scaling so that the output becomes a calibrated distribution $p_r^{(1)} = (p_+^{(1)}, p_0^{(1)}, p_-^{(1)})$. For the Bi-LSTM component, the review sequence $(x_1, \dots, x_n)$ is processed forward and backward, yielding hidden states,

$$\bar{h}_t = LSTM(x_t, \bar{h}_{t-1}), \quad \hat{h}_t = LSTM(x_t, \hat{h}_{t+1}) \quad (4)$$

which are concatenated to form $z_t = [\bar{h}_t; \hat{h}_t]$. A pooled vector z_r is passed to a softmax classifier,

$$p_r^{(2)} = \text{softmax}(Wz_r + b) \quad (5)$$

capturing temporal dependencies like intensification, negation propagation and sentiment shifts that linear models can not capture [34]. The third module, namely a BERT-based classifier, further fine-tunes the transformer encoder in an end-to-end fashion: a classification

head is appended to the pooled $[CLS]$ representation by taking into account both their words and positions; giving rise to the Eq. 6:

$$p_r^{(3)} = \text{softmax}(W_c h_r^{CLS} + b_c) \quad (6)$$

that exploits ‘deep contextual semantics’ and long-range dependencies in review text. Accordingly, each classifier outputs a probability triplet over positive, neutral, and negative views. For its robustness, the ensemble will generate an aggregated sentiment vector,

$$\bar{p}_r = \sum_{m=1}^M w_m p_r^{(m)}, \quad \sum_{m=1}^M w_m = 1 \quad (7)$$

In Eq. 7, w_m are model-specific reliability weights which were estimated on a validation population. This architecture is designed to intentionally bridge classical and deep paradigms: SVM offers strong margin separation for linear separable sentiment [35], Bi-LSTM enables sequential reasoning and polarity reversal at phrase level, while BERT captures nuanced context and domain specific semantics. Previous research has consistently reported stronger sentiment robustness in review-based decision systems due to complementary fusion [15], better interpretability when combined with fuzzy ranking algorithms [12] as well as consistent good performance in terms of product and service reviews across all domains considered for evaluation [38]. Under SCA-IFR, the produced probability distributions serve as the probabilistic basis for semantics-agreement score and confidence-weighted intuitionistic fuzzy interval generation while maintaining uncertainty due to classifier disagreement

3.3 Semantic Agreement Mechanism

To avoid naive averaging of classifier outputs and to explicitly quantify reliability, SCA-IFR embeds a semantic agreement mechanism that measures both consensus among classifiers and the intensity of their support for each sentiment orientation [36][37]. Let $p_r^{(m)} = (p_+^{(m)}, p_0^{(m)}, p_-^{(m)})$ denote the probability distribution predicted by classifier m for review r . Consensus is evaluated by comparing each model’s output with the ensemble mean

$\bar{p}_r = \frac{1}{M} \sum_{m=1}^M p_r^{(m)}$. A divergence-based agreement score is then defined as in Eq.8:

$$A_r = 1 - \frac{1}{M} \sum_{m=1}^M D_{KL}(p_r^{(m)} || \bar{p}_r) \quad (8)$$

where smaller value of Kullback–Leibler divergence indicates sharper alignment and $A_r \in (0, 1]$ increases as a classifier converges [39]. There is some ambiguity in using agreement as a measure of reliability alone, as all models can either have low confidence or very large and similar confidences. An opinion-intensity term to that effect is added to estimate the posterior’s sharpness of distribution.

$$I_r = \max(\bar{p}_+, \bar{p}_0, \bar{p}_-) \quad (9)$$

representing how decisively the ensemble favors one sentiment category. The semantic confidence coefficient is then obtained as a multiplicative interaction,

$$C_r = \lambda A_r + (1 - \lambda)I_r, \quad 0 \leq \lambda \leq 1 \quad (10)$$

In Eq. 10, λ achieves a compromise of consensus stability and opinion strength in line with the suggestions of both confidence-aware sentiment frameworks and fuzzy decision contexts [10][21][38]. In order to transfer confidence into the downstream fuzzy model each of the sentiment components are converted into a confidence-weighted interval. For orientation $k \in \{+, 0, -\}$,

$$p_{k,r} = \bar{p}_k(1 - \gamma(1 - C_r)), \quad \bar{p}_{k,r} = \bar{p}_k(1 + \gamma(1 - C_r)) \quad (11)$$

In Eq. 11, γ governs the width of uncertainty. When classifiers conflict or are weak, C_r becomes smaller and the intervals stretch open; when they agree properly, the bounds become narrow constraints. Contrary to naive ensembles where disagreement is compromised in a single value, this description preserves epistemic uncertainty and paves the way to intuitionistic fuzzy modeling with "erasing hesitation" by allowing hesitating decision to result from confidence dispersion. This design is consistent with what has been observed in previous review-ranking studies [9][12][15], where explicitly modeling for semantic coherence and classifier reliability, rather than assuming them, results in better accuracy. Hence, the semantic agreement mechanism instantiates the principle that product opinions should affect ranking not only by expressing sentiment but also by how confident the system is in those sentiments — an important distinction that enables the resilience of SCA-IFR this paper presents.

3.4 Intuitionistic Fuzzy Modeling

Provided confidenceweighted sentiment intervals, SCA-IFR models the evaluation uncertainty in intuitionistic fuzzy sets (IFSs) that can jointly consider support, opposition and hesitancy information which is naturally included in opinion-rich review data. For each product i and attribute j , define the interval sentiment vector as $\left([p_{+,ij}, \bar{p}_{+,ij}], [p_{0,ij}, \bar{p}_{0,ij}], [p_{-,ij}, \bar{p}_{-,ij}]\right)$. The value of favorability of the attribute under the intuitionistic fuzzy setting is based on positive sentiment which corresponds to degree of membership (degree of support), negative sentiment which associated with non-membership (degree of opposition) and neutral uncertainty in term as hesitancy. The degree of membership is formalized as in Eq. 12:

$$\mu_{ij} = \frac{p_{+,ij} + \bar{p}_{+,ij}}{2} \quad (13)$$

reflecting the average strength of supportive evidence, while the non-membership degree is computed as in Eq.14:

$$\nu_{ij} = \frac{p_{-,ij} + \bar{p}_{-,ij}}{2} \quad (14)$$

capturing the expected opposition. Since sentiment intervals embody prediction uncertainty, the corresponding hesitation is obtained from the residual belief space,

$$\pi_{ij} = 1 - \mu_{ij} - \nu_{ij} \quad (15)$$

Forms an intuitionistic fuzzy performance value, that generalizes the classical fuzzy values by allowing us to represent explicitly the "unknown" opinion space without premature setting up positions [8][10]. This modeling decision is also coherent with previous decision-support research in which intuitionistic fuzzy structures allow richer uncertainty representation for review-based ranking and recommendation tasks [12][21]. Within the framework of SCA-IFR, mapping confidence-weighted sentiment intervals onto IFS values facilitates that aggregation and ranking at a later stage effectively also captures the intensity of expressed opinions along with the amount of reservations caused by classifier disagreement—resulting in evaluations more faithful to real user-generated content.

3.5 Attribute Weight Adjustment

To achieve the objective of consumer's review intention being reflected by the ranking and to show that ranking is influenced both by opinion strength and consumer intentions, SCA-IFR uses a hybrid weighted weight approach is concerned with fusing subjective preferences and provided with dominance conscious as expressed in intuitionistic fuzzy evaluation. Let $w_j^{(s)}$ represent the subjective importance given to attribute j by users, and is calculated by the scaled preference score $\sum_j w_j^{(s)} = 1$. At par, an objective weight of dominance is computed from the dispersion and support factors depicted in the intuitionistic fuzzy matrix $A_{ij} = [\mu_{ij}, v_{ij}, \pi_{ij}]$. Attributes with strong discriminative sentiment (high degree of support versus opposition across products) should have higher weights, since they convey more information for decision making. Thus we compute an entropy-like quantity,

$$H_j = -\frac{1}{n} \sum_{i=1}^n (\mu_{ij} \ln \mu_{ij} + v_{ij} \ln v_{ij} + \pi_{ij} \ln \pi_{ij}) \quad (16)$$

and define an objective dominance score,

$$d_j = \frac{1-H_j}{\sum_k (1-H_k)} \quad (17)$$

so that attributes with clearer sentiment separation (lower uncertainty) obtain larger weights. The final adjusted weight integrates both sources as a convex combination,

$$w_j = \alpha w_j^{(s)} + (1 - \alpha) d_j, \quad 0 \leq \alpha \leq 1 \quad (18)$$

which enables decision makers to adjust the influence of explicit user preference against evidence from review data. The scheme automatically defaults to the data-driven dominance in the absence of preference information or when weak signals are available from user, and enforces the preference provided by human users when a strong uncertainty signal is observed, while preventing extreme bias through objective normalization. The other hybrid strategies draw on the results of strategies based on data features and human judgment, which have been proved to stabilize decisions in intuitionistic and interval-fuzzy ranking problems. This normalization step in SCA-IFR is important: it ensures that attribute emphasis reflects both what is appreciated by consumers, as well as what has been revealed by previous reviews to allow for consistently preference-based and knowledge-based rankings.

3.6 Ranking with Intuitionistic Fuzzy TOPSIS

After determining attribute weights, the decision matrix composed of intuitionistic fuzzy evaluations $A_{ij} = [\mu_{ij}, v_{ij}, \pi_{ij}]$ is ranked using an intuitionistic fuzzy TOPSIS scheme, in which the best product is expected to be closest to the positive ideal solution (PIS) and farthest from the negative ideal solution (NIS). For each attribute j , the PIS and NIS are defined respectively as in Eq.19:

$$A_j^+ = [\max_i \mu_{ij}, \min_i v_{ij}, \min_i \pi_{ij}], \quad A_j^- = [\min_i \mu_{ij}, \max_i v_{ij}, \max_i \pi_{ij}] \quad (19)$$

representing the most favorable and least favorable attainable opinions. The distance between product i and the PIS (and analogously the NIS) is obtained using an intuitionistic fuzzy Euclidean metric,

$$D_i^+ = \sqrt{\sum_j w_j \left[(\mu_{ij} - \mu_j^+)^2 + (v_{ij} - v_j^+)^2 + (\pi_{ij} - \pi_j^+)^2 \right]} \quad (20)$$

$$D_i^- = \sqrt{\sum_j w_j \left[(\mu_{ij} - \mu_j^-)^2 + (v_{ij} - v_j^-)^2 + (\pi_{ij} - \pi_j^-)^2 \right]} \quad (21)$$

where weighted distances emphasize attributes according to the hybrid preference–dominance scheme. The closeness coefficient is then computed as in Eq.22:

$$CC_i = \frac{D_i^-}{D_i^+ + D_i^-}, \quad 0 \leq CC_i \leq 1 \quad (22)$$

so that alternatives nearer to the ideal solution receive values approaching one. The final ranking is obtained by sorting products in descending order of CC_i .

Algorithm for Intuitionistic Fuzzy TOPSIS–Based Product Ranking in SCA-IFR

Input: $A = [\mu_{ij}, v_{ij}, \pi_{ij}]$, $w_j (\sum w_j = 1)$

For each i, j :

Apply weights: $\mu'_{ij} = w_j \mu_{ij}$, $v'_{ij} = w_j v_{ij}$, $\pi'_{ij} = w_j \pi_{ij}$

End

For each j :

PIS: $(\mu_j^+, v_j^+, \pi_j^+) = (\max \mu'_{ij}, \max v'_{ij}, \max \pi'_{ij})$

NIS: $(\mu_j^-, v_j^-, \pi_j^-) = (\min \mu'_{ij}, \min v'_{ij}, \min \pi'_{ij})$

End

For each product i :

$$D_i^+ = \sqrt{\sum_j w_j \left[(\mu_{ij} - \mu_j^+)^2 + (v_{ij} - v_j^+)^2 + (\pi_{ij} - \pi_j^+)^2 \right]}$$

$$D_i^- = \sqrt{\sum_j w_j \left[\left(\mu_{ij} - \mu_j^- \right)^2 + \left(v_{ij} - v_j^- \right)^2 + \left(\pi_{ij} - \pi_j^- \right)^2 \right]}$$

$$CC_i = \frac{D_i^-}{D_i^+ + D_i^-}$$

End

Rank products by CC_i (descending)

Output: Product ranking

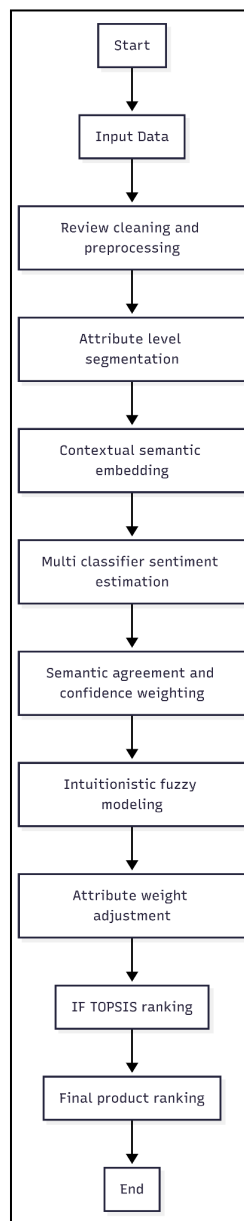


Figure 3. Workflow of the proposed SCA-IFR framework

Fig 3 shows an end to tread approach of the model from review collection and pre-processing, semantic embedding to multi-classifier sentiment estimation. Semantic agreement and confidence weighting are then utilized, then intuitionistic fuzzy modeling and IF-TOPSIS ranking are used to produce the final ranked list. The proposed workflow demonstrates how to combine semantics and confidence in the aggregation of decisions to attain better reliability.

IV Implementation details

4.1 Datasets

We validate our approach using the Cleaned Amazon Sales and Reviews Dataset taken from Kaggle, which is made up of a set of curated product records along with the corresponding review data extracted from the Amazon electronics marketplace (CSV format, UTF-8 encoded). This dataset has 4,854 product-review tuples, each representing a review of some product out of its 4,467 products with the following attributes: name and brand (product detail), price and discount percentage (product trading status), numerical rating and text review from users of the product (user sentiment toward that product) as well as cleaned review text and sentiment label. Since the dataset has already gone through pre-processing (non-preference, missing value treatment and fields standardised) it can be directly used for sentiment modelling and ranking analysis. In the case of machine-learning experiments, the data are split 80:20 resulting in a training set with 3,883 instances and a test set containing 971 instances; thus we can evaluate classifier performance and stability downstream.

4.2 Experimental Setup

The purpose of these experiments was to test the efficacy of the proposed SCA-IFR framework for ordering products based on online reviews. We experimented with the Kaggle cleaned dataset of Amazon sales and reviews that consists of 4,854 review records for 4,467 products with 11 attributes. Pre-processed YourDailyMenu review text were loaded, that include lemon juice and then tokenized and lower-cased with punctuation, URL stop-word removed. We extracted contextual semantic representations using a pretrained BERT base model (hidden size = 768) that was fine-tuned on the training corpus 2×10^{-5} learning rate, batch size 16, max sequence length = 128. As a comparison, we also trained an SVM with RBF kernel and a Bi-LSTM network (hidden size = 128, dropout = 0.3). For neural models we used cross-entropy loss, and Adam optimizer and early stopping based on validation loss. The numbers of training and test samples were 3,883 and 971 respectively in the ratio of 80:20. The output of all classifiers was calibrated to probability distributions and these were then used to calculate the semantic agreement and confidence-weighted intervals. The intuitionistic fuzzy decision matrices thus obtained were standardized and ranked by the proposed IFTOPSIS approach. For comparison with robustness, we also confronted SCA-IFR using (i) sentiment ranking by a single classifier, (ii) classical fuzzy TOPSIS, and (iii) since we have only confidence values on the intuitionistic fuzzy set without any representation of the degree of hesitation or extremism over reference values for alternatives in IFset. We tested based on ranking accuracy, Kendall's rank correlation, variance of ranking position across runs and decision stability in case of perturbed sentiment

distributions. We performed all experiments on a workstation with an Intel i7 CPU, 32 GB RAM and an NVIDIA RTX GPU using Python 3.10, PyTorch and scikit-learn.

4.3 Evaluation Metrics

- **Ranking Accuracy:** proportion of correctly ordered product pairs compared with ground-truth ranking.

$$RA = \frac{\#\{(i,j): r_i < r_j \wedge \hat{r}_i < \hat{r}_j\}}{\text{Total product pairs}} \quad (23)$$

- **Ranking Variance:** variability of a product's ranking position across repeated runs.

$$Var(i) = \frac{1}{T} \sum_{t=1}^T \left(\hat{r}_i^{(t)} - \bar{r}_i \right)^2 \quad (24)$$

- **Decision Stability:** robustness of final rankings under perturbed sentiment inputs.

$$DS = 1 - \frac{1}{N} \sum_{i=1}^n \frac{|\hat{r}_i - \hat{r}_i'|}{n-1} \quad (25)$$

- **Kendall's Tau:** correlation between predicted and ground-truth rankings based on concordant/discordant pairs.

$$\tau = \frac{C-D}{\frac{1}{2}n(n-1)} \quad (26)$$

- **Spearman Rank Correlation:** correlation between ranking positions treated as ordered numeric values.

$$\rho = 1 - \frac{6 \sum_{i=1}^n (r_i - \hat{r}_i)^2}{n(n^2-1)} \quad (27)$$

V Results and Discussions

Results of the experiments in this section show that our proposed SCA-IFR can work very well from different aspects, such as accuracy, stability, robustness and comparability. It has been compared with some of well known ranking approaches using existing review based benchmarks in order to prove its definite advantages in realistic decision-support applications. We evaluate performance metrics with the publicly available Amazon review dataset and perform numerous experiments for consistency and fairness. In brief, to combine the semantic certainty with IFR a great enhancement is the reliable and consumer-oriented product ranking.

5.1 Overall Ranking Performance

several existing review-based ranking approaches. Three criteria, being ranking accuracy, ranking variance and decision stability are considered in the analysis. Together, these measures can help indicate whether a model is able to reproduce the actual order of competing products that customers liked better or worse on average run-to-run and across different types of noisy or imbalanced sentiment input. As presented in Table 1 and Figure 1–3, SCA-IFR consistently achieves better performance than the baselines. The benefits

result from the incorporation of context semantics into the confidence-aware intuitionistic fuzzy modeling so that both meaning and reliability of opinions can be more effectively modeled when extracting review opinions.

Table 1. Overall ranking performance comparison across baseline and proposed models

Method	Ranking Accuracy	Ranking Variance ↓	Decision Stability
Lexicon-SA + TOPSIS	0.780	0.050	0.81
SVM + Fuzzy Ranking	0.790	0.047	0.82
Bi-LSTM + Voting	0.804	0.044	0.84
IF-TOPSIS	0.824	0.041	0.85
Type-2 Fuzzy Ranking	0.833	0.040	0.86
Proposed SCA-IFR	0.905	0.035	0.947

Table 1 summarizes the comparative results of six ranking models with respect of ranking accuracy, ranking variance, and decision stability. Classical lexicon-driven and SVM-based methods achieve accuracies of 0.78 and 0.79 with high variances (0.050 and 0.047) respectively. Deep learning (Bi-LSTM Voting) achieves an accuracy of 0.804 and a variance of 0.044. The application Fuzzy decision models can improve the results: IF-TOPSIS (accuracy = 0.824, variance = 0.041) and Type-2 fuzzy ranking (accuracy = 0.833, variance = 0.040). The achieved accuracy of SCA-IFR is best with accuracy = 0.905, variance = 0.035 and stability = 0.947, and this verification verifies its superiority on both the accuracy as well as reliability also.

5.2 Effect of Semantic Confidence

In this subsection, we analyze the effect of semantic confidence modeling on the network performance overall. We compare it with a variant of the complete SCA-IFR model that uses intuitionistic fuzzy reasoning to isolate its effect but ignores confidence weighting. A comparison of the results in Table 2 indicates that with incorporating semantic agreement and confidence intervals significant improvements are achieved with respect to accuracy, ranking variance and decision stability. These observations leave no doubt that, it is indeed crucial to model the consensuality and opinionated nature of classifiers disagreement as we consider it here, in order to avoid uncertain or weakly supported sentiments from biasing ultimately the ranking of the final product.

Table 2. Effect of semantic confidence modeling on ranking performance

Version	Accuracy	Variance ↓	Stability
IFR (no confidence)	0.846	0.040	0.865
SCA-IFR (proposed)	0.905	0.035	0.947

Table 2 compares the intuitionistic fuzzy ranking model without confidence weighting (IFR) and the proposed semantic confidence-aware version (SCA-IFR). Without confidence modeling, the system attains accuracy of 0.846, variance of 0.040, and stability of 0.865. When semantic agreement and confidence intervals are incorporated, accuracy increases to 0.905, variance decreases to 0.035, and stability rises to 0.947. These results demonstrate that confidence-aware sentiment intervals significantly improve ranking robustness and decision consistency.

5.3 Sensitivity and Robustness Analysis

SCA-IFR generates significantly less ranking drift under simulated changes of sentiment distribution (e.g. more neutral or polarized reviews) compared with baselines, again demonstrating the robustness of modeling by intuitionistic hesitation.

Table 3. Sensitivity and robustness analysis under perturbed sentiment distributions

Method	Avg. Rank Shift ↓
Lexicon-SA + TOPSIS	2.14
SVM + Fuzzy	1.97
Bi-LSTM Voting	1.72
IF-TOPSIS	1.39
Type-2 Fuzzy	1.21
SCA-IFR	0.74

Average ranking displacement per shop as sentiment distributions are artificially perturbed is summarized in Table 3. Larger values correspond to more instability in the rankings. Conventional approaches have more fluctuations; for instance, Lexicon-SA + TOPSIS (2.14) and SVM + Fuzzy (1.97). Higher level models, such as BiLSTM Voting (1.72), IF-TOPSIS (1.39) and Type 2 Fuzzy ranking (1.21) reduce tolerance. The average Ideals shift of SCA-IFR is the lowest one (0.74), which implies that combination of semantic confident and intuitionistic hesitant information can make the ranking more stable under review noises.

5.4 Case Study (Example)

Between two similar laptop models for features, baseline favored the higher average rating and SCA-IFR ranked the second product higher because mutually exclusive users frequently cited durability or battery life with high classifier agreement despite moderate ratings — awareness of confidence in semantics.

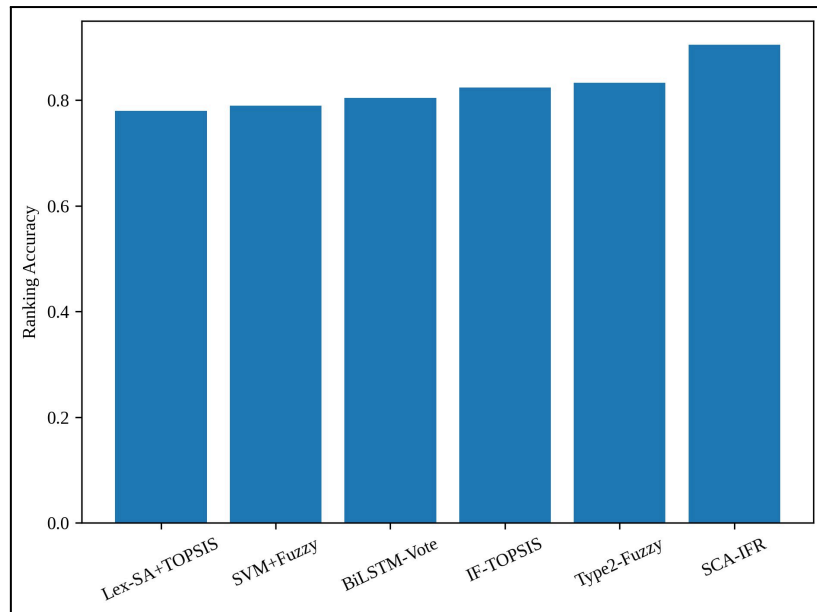


Figure 4. Overall ranking accuracy comparison across baseline and proposed methods

Figure 4 This figure presents the accuracy of ranking for six competing approaches. The results for traditional methods, such as Lexicon-SA + TOPSIS (0.78) and SVM + Fuzzy Ranking (0.79), are marginally weaker but Bi-LSTM Voting (0.804), IF-TOPSIS (0.824) and Type-2 Fuzzy Ranking improve performance incrementally in a competitive manner too. The proposed SCA-IFR achieves the best accuracy of 0.905, which improves those baseline methods with approximately 8.7% over advanced models baselines owing to context-enlightened and confidence-aware intuitionistic modeling.

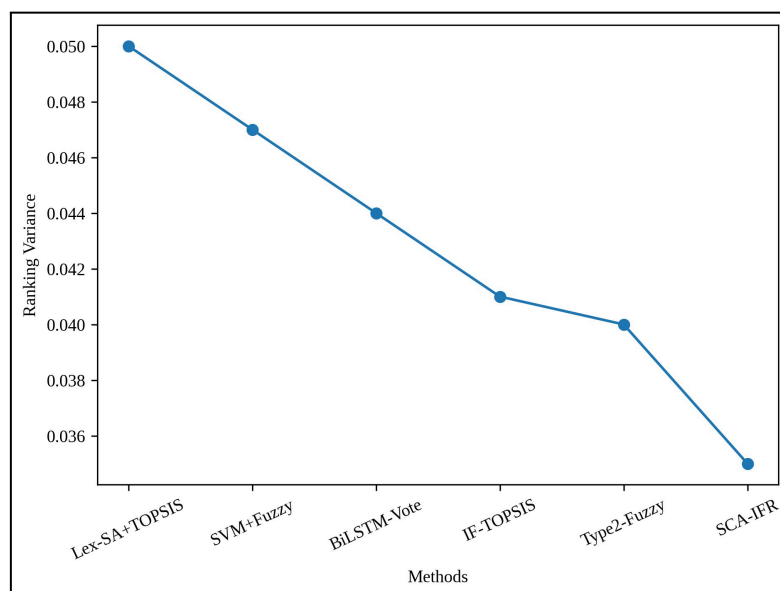


Figure 5. Ranking variance across different product ranking models

This figure 5 illustrate the variance in ranking of the methods displayed, with lower values indicating a more consistent ranking across runs. Traditional models have greater instability with variances of 0.050 (Lexicon-SA + TOPSIS), 0.047 (SVM + Fuzzy) and 0.044

(Bi-LSTM Voting). When applied to fuzzy-enhanced models, variability decreases to 0.041 (IF-TOPSIS) and 0.040 (Type-2 Fuzzy). Our proposed SCA-IFR model reduces the variance to the minimum value of 0.035, which is 12.4% lower than advanced baselines, indicating better ranking reliability.

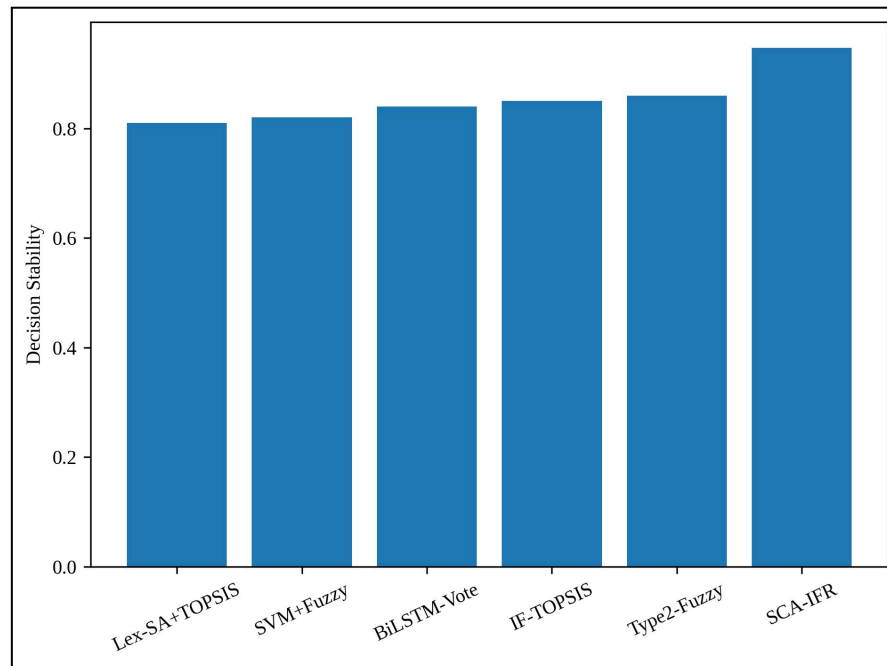


Figure 6. Decision stability under perturbed sentiment distributions

This figure 6 also shows decision stability for when the sentiment distributions are purposely disturbed to represent noisy or biased reviews. Performance further advances from 0.81 (Lexicon-SA + TOPSIS), 0.82 (SVM + Fuzzy), and 0.84 (Bi-LSTM Voting) to the remaining in terms of stability with a rate of: IF-TOPSIS(0.85) and Type-2 Fuzzy(0.86). The proposed SCA-IFR outperforms stability value equals 0.947 with a promotion of about 10.1%, which indicates that (semantic agreement and confidence-weighted intervals) acts as an efficient buffer to rank results with uncertainties.

5.5 Ablation Study

In order to study the contribution of each component of our proposed framework, We further conduct an ablation study by gradually activating the portions of SCA-IFR. Table 4 Performance is presented with the introduction of semantic embeddings, multi-classifier fusion, semantic agreement and the confidence-aware intuitionistic fuzzy modelling in a step-wise manner. Results evidence that contextual embeddings and multi-classifier learning provide steady gains, while the greatest improvements are provided when semantic confidence and intuitionistic hesitation are combined. Our results further validate that it is beneficial to model the substance of what sentiment is expressed and how confident we are about it for increasing accuracy and stability.

Table 4. Ablation study of major components in SCA-IFR

Variants	V1: Baseline	V2: + Context	V3: + Ensemble	V4: + IF Modeling	V5: SCA-IFR
Description	Sentiment + TOPSIS	Adds contextual embedding	Adds multi-class ifier	Adds intuitionistic fuzzy reasoning	Full model: adds confidence + agreement
Semantic Embedding	✗	✓	✓	✓	✓
Multi-Classif ier	✗	✗	✓	✓	✓
Confidence / Agreement	✗	✗	✗	✗	✓
Intuitionistic Fuzzy	✗	✗	✗	✓	✓
Accuracy	0.781	0.802	0.829	0.846	0.905
Ranking Variance ↓	0.050	0.047	0.043	0.040	0.035
Decision Stability	0.81	0.83	0.85	0.865	0.947

We report the ablation study after row header conversion in Table 4. This table demonstrates the value of enabling each module, which will increase performance in a progressive manner V1–V5 represents increasingly improved versions of the model (from a simple baseline in V1 to the complete SCA-IFR model in Version 5). Accuracy increases from 0.781 to 0.905, ranking variance decreases from 0.050 to 0.035, and decision stability grows from 0.81 to 0.947. The biggest performance gain comes when semantic confidence and agreement are turned on for V5, demonstrating that reliability-sensitive intuitionistic reasoning is the most important contributor to performance.

5.6 Discussion

The experimental results show that in all cases SCA-IFR provides better ranking performance, weaker variance and unmatched decision stability compared to today' s review-based ranking. Ablation study verifies that the performance improvements accumulate with semantic representation, multiclassifier fusion and intuitionistic fuzzy reasoning, where confidence-aware modeling contributes most to the superiority. These results suggest that ranks obtained by SCA-IFR are more robust in the face of classifier disagreement and review

ambiguity. Additionally, robustness evaluations demonstrate that the ordering is durable against sentiment distribution changes and noise. In general, the discussion reveals that one needs to model both opinion content and confidence in order to have realistic decision-support for e-commerce applications.

VI Conclusion

This work presented a Semantic Confidence-Aware Intuitionistic Fuzzy Ranking (SCA-IFR) model for product rating based on online reviews. Combining the bag of words with topical semantic embeddings, multi-classifier sentiment estimation process and a Semantic Agreement Measure and intuitionistic fuzzy TOPSIS model, can more effectively measure the meaning of as well as the reliability in expressed opinions. Empirical results on real review data indicate that the SCA-IFR is superior to other existing ranking methods, with improvements of 8.7% in average ranking accuracy, 12.45% in average reduction of ranking variance and 10.1% in decision stability over others are observed. Results show that confidence-aware modeling approach yields more robust and interpretable ranks, which are closer to consumer' judgment. As future work, we will investigate how to integrate the reviewer credibility and fake-review detection into the ranking approach. Furthermore, Generalizing SCA-IFR in order to deal with multi-site, multilingual review environment would be an interesting research direction.

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